

Package ‘micsPlusTableR’

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Title MICS Plus Excel Tabulation and Review Application

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Description Runs interactive and scriptable workflows for preparing, tabulating, checking, reviewing, and exporting MICS Plus tables from an Excel tabulation plan, survey data files, and a survey-specific preparation script.

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micsPlusTableR-package

micsPlusTableR: MICS Plus Excel tabulation and review

Description

Prepare, tabulate, check, review, and export MICS Plus survey tables using an Excel tabulation plan. Work interactively with `run_app()` or use the scripting functions with an isolated `mics_session()`.

Inputs

Supply an Excel tabulation plan with an IDX worksheet and a worksheet for each table, household and household-member SPSS files, and a survey-specific R preparation script that reads `hh_path` and `hl_path` and creates data frames named `hh` and `hl`. A sibling `FIES-inputs` directory is optional.

User setup

Install R and an editor such as RStudio Desktop. Before installing this package or starting the app, create or open your own survey project. In RStudio, use File > New Project for a new or existing folder, or File > Open Project for an existing `.Rproj` file. Install the package with `remotes::install_github("barisr/micsPlusTableR")` after installing `remotes`, then restart R. The bundled `doc/user-guide.pdf` and `doc/user-guide.html` explain computer setup, project creation, installation, and file uploads for beginners. The app's User's Guide tab opens the guide after launch.

Positron, VS Code, Cursor, Jupyter with an R kernel, and R's own console can also run the package. Configure R support and start R in the survey folder. Check `getwd()` and `here::here()` or set `output_dir` explicitly; use `launch.browser = TRUE` if needed. Remote sessions need browser routing and access to server files. Users do not need the maintainer's project.

Interactive workflow

Reopen the survey project on each visit; installation is normally needed only once per R library. Select inputs again for each app session. Start `run_app()`. In Download Files, load the list and download one, several, or all survey folders to the current project or a chosen location. This saves plans and preparation materials; household/member data must be obtained from <https://mics.unicef.org/surveys> with permission. In Data Preparation, select the survey and wave, provide the inputs, prepare the data, calculate tables, review checks, and export the results. The app creates `micsPlusTableR-output` in the project root by default. To select another location, pass the `output_dir` argument to `run_app()`.

Scripting workflow

The main steps are:

- `mics_session()` creates an isolated session.
- `prepare_mics_data()` runs the preparation script.
- `tabulate_mics_table()` calculates cell results.
- `check_mics_table()` checks consistency.
- `pivot_mics_table()` arranges results for review.
- `create_excel_workbooks()` creates output workbooks.
- `write_mics_table()` writes the results.
- `write_mics_footnotes()` writes table footnotes.
- `read_previous_mics_table()` reads an earlier output.
- `compare_mics_tables()` compares two outputs.

See [offline-workflow](#) for a complete scripting example.

Survey and wave configuration

Maintain `inst/extdata/survey_choices.csv` in the package source. It remains a plain, editable CSV on GitHub and is bundled automatically on installation. See [survey_choices](#) for its columns, validation rules, and update workflow.

Outputs and interpretation

Outputs include Excel workbooks and long-format observations. The app offers Excel, CSV, and RDS downloads of the full prepared long-format dataset. Use `as_sdmx_compatible()` to convert observations to the project's SDMX-compatible contract. Use `validate_sdmx_compatible()` to check them. This contract does not by itself establish compliance with an external SDMX data structure. Table checks assess consistency; analysts remain responsible for preparation scripts, survey design, denominators, and interpretation of estimates.

Author(s)

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See Also

[run_app\(\)](#), [survey_choices](#), [offline-workflow](#)

Examples

```
session <- mics_session()
session
system.file("extdata", "survey_choices.csv", package = "micsPlusTableR")
```

as_sdmx_compatible	Add an SDMX-compatible observation profile to MICS cell results
--------------------	---

Description

Converts either the package's engine cell results or an already standardized MICS cell table into a 35-variable long-format contract adapted from the mics-translation project. The output omits the custom table_id, value, estimate, digits, and unw duplicates; use TABLE_ID, OBS_VALUE, DECIMALS, and denominator_n instead. Input fields are unchanged. Source coordinates remain in row_order and column_order; the redundant row_id, column_id, ROW_ID, and COLUMN_ID fields are omitted. The output also omits unweighted_n, which duplicates denominator_n for engine cells. For standardized input, the supplied denominator_n is retained. The optional input unit_measure override is applied to UNIT_MEASURE but is not included as a separate output field.

Usage

```
as_sdmx_compatible(cells, config, table_id = NULL, table_context = NULL)
```

Arguments

cells	A data frame with one row per table cell.
config	A list containing frequency, ref_area, survey_id, and time_period.
table_id	Table identifier. Required for raw engine cell results.
table_context	Optional parsed tabulation context used to recover multi-level column headers and suppression settings.

Value

A tibble with one row per observation and the 35 standardized fields.

check_mics_table	Run all applicable consistency checks
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Description

Run all applicable consistency checks

Usage

```
check_mics_table(session, table = NULL, tolerance = 1e-06)
```

Arguments

session	A tabulation session.
table	Cell results; defaults to the session's latest result.
tolerance	Maximum absolute difference treated as consistent.

Value

A list containing a summary data frame and detailed check tables.

compare_mics_tables	<i>Compare two generated tables</i>
---------------------	-------------------------------------

Description

Compare two generated tables

Usage

```
compare_mics_tables(
  previous,
  current,
  ignore_col_names = FALSE,
  show_diff_over = 0
)
```

Arguments

previous	Previous or approved table.
current	Current table.
ignore_col_names	Compare columns by position instead of name.
show_diff_over	Minimum absolute numeric difference to report.

Value

A data frame containing differing cells.

create_excel_workbooks	<i>Create output workbooks from a tabulation plan</i>
------------------------	---

Description

Copies the tabulation plan to standard output names without overwriting an existing file. A numeric suffix is added after a filename collision.

Usage

```
create_excel_workbooks(
  path_tab_excel,
  output_dir,
  country_code,
  period,
  wave,
  target = c("both", "output", "formatted"),
  time_tag = format(Sys.time(), "%Y%m%d%H%M%S")
)
```

Arguments

path_tab_excel	Existing .xls or .xlsx tabulation plan.
output_dir	Directory where workbooks will be created.
country_code	Short country code used in filenames.
period	Survey period used in filenames.
wave	Wave label used in filenames.
target	One of "output", "formatted", or "both".
time_tag	Timestamp used in filenames.

Value

A list with output and formatted paths.

list_survey_files	<i>List or download survey tabulation plans and preparation files</i>
-------------------	---

Description

Discover or download tabulation plans, preparation scripts, and their supporting reference assets from the separate survey files repository. Household and household-member microdata are not distributed here; obtain them separately from the MICS Plus website with the required permission. The repository must be public. No GitHub account or token is needed for these plans and preparation files.

Usage

```
list_survey_files(repo = "barisr/micsPlusTableR-Files", ref = "main")

download_survey_files(
  surveys = NULL,
  files = NULL,
  dest_dir = "inputs",
  overwrite = FALSE,
  quiet = FALSE,
  repo = "barisr/micsPlusTableR-Files",
  ref = "main"
)
```

Arguments

repo	GitHub repository in owner/name form.
ref	Branch, tag, or commit to download. Defaults to "main".
surveys	Character vector of root survey folder names, such as "JAM_W1". NULL selects all surveys.
files	Character vector of exact repository-relative paths from list_survey_files() . NULL selects all files in the selected surveys.
dest_dir	Destination folder, relative to the current R working directory unless an absolute path is supplied. Defaults to "inputs".

overwrite	Replace existing files? Defaults to FALSE; any existing selected file stops the operation before downloading file contents.
quiet	Suppress file download progress and the completion message?

Details

Survey folders use a three-letter country code and wave, such as JAM_W1. Downloads include root-level Excel plans (.xls, .xlsx) and .R scripts or .RData reference assets inside prep-files-* subfolders. Other files, including SPSS microdata and hidden files, are excluded. Preparation subfolders are preserved. A single Git commit is resolved before listing or downloading, so changes to the branch during a download cannot mix remote versions. Newly published survey folders are discovered without a package update.

Downloads are staged and their sizes checked before destination files are written. A network failure leaves existing destination files unchanged. Files are downloaded in binary mode and are never executed. Selecting individual files may omit preparation helpers; select a whole survey folder for all its published materials. Microdata must still be obtained separately. Git LFS pointers are not supported.

Internet access is required. Unauthenticated GitHub API rate limits apply; if a request fails, check the connection and public repository/ref, or retry later. The browser ZIP link in the user's guide is an alternative.

Value

list_survey_files() returns a data frame with survey, path, size (bytes), and url columns, plus a commit attribute. download_survey_files() invisibly returns the selected rows with an additional local_path column and the same commit attribute.

Examples

```
## Not run:
available <- list_survey_files()
unique(available$survey)
download_survey_files("JAM_W1")
download_survey_files(c("JAM_W1", "MNG_W2"), dest_dir = "survey-inputs")
download_survey_files() # Plans and preparation files for all surveys
available$path # Copy exact paths to select individual files
download_survey_files(files = available$path[1])

## End(Not run)
```

mics_session

Create a non-Shiny MICS Plus tabulation session

Description

Create a non-Shiny MICS Plus tabulation session

Usage

```
mics_session(hh = NULL, hl = NULL)
```

Arguments

- hh Optional prepared household data frame.
- hl Optional prepared household-member data frame.

Value

A mutable, isolated tabulation session environment.

mics_table_name	<i>Get the current table title</i>
-----------------	------------------------------------

Description

Get the current table title

Usage

```
mics_table_name(session)
```

Arguments

- session A session with a tabulation sheet already read.

Value

The table title as a character scalar.

offline-workflow	<i>Run MICS Plus tabulation workflows outside Shiny</i>
------------------	---

Description

These functions provide the supported non-Shiny interface to the packaged tabulation engine. The explicit session keeps survey data, plan context, extra tables, and calculated results together without writing them into the user’s global workspace.

Details

Prepare survey data in a new session, calculate a table from the Excel plan, check its consistency, and pivot it for review before writing results. The example requires analyst-supplied files. See the package overview for links to the function references and workbook export steps.

See Also

[micsPlusTableR-package](#)

Examples

```
## Not run:
session <- mics_session()
prepare_mics_data(
  session,
  "hh.sav",
  "hl.sav",
  prep_script = "survey-preparation/survey_prep.R",
  output_dir = "micsPlusTableR-output"
)
result <- tabulate_mics_table(session, "TabulationPlan.xlsx", "1.1")
checks <- check_mics_table(session, result)
preview <- pivot_mics_table(session, result, type = "header")

## End(Not run)
```

pivot_mics_table	<i>Pivot long-format cell results for review</i>
------------------	--

Description

Pivot long-format cell results for review

Usage

```
pivot_mics_table(
  session,
  table = NULL,
  formatted = FALSE,
  type = c("index", "header", "logic")
)
```

Arguments

session	A tabulation session.
table	Cell results; defaults to the session's latest result.
formatted	FALSE, TRUE, or "view" as supported by the engine.
type	One of "index", "header", or "logic".

Value

A wide review table.

prepare_mics_data	<i>Run a survey preparation script outside the Shiny application</i>
-------------------	--

Description

Run a survey preparation script outside the Shiny application

Usage

```
prepare_mics_data(session, hh_path, hl_path, prep_script, output_dir = NULL)
```

Arguments

session	A session created by <code>mics_session()</code> .
hh_path	Path to the household .sav file.
hl_path	Path to the household-member .sav file.
prep_script	Path to the trusted preparation .R script. A sibling FIES-inputs folder is detected automatically.
output_dir	Main output folder. If the script's folder contains FIES-inputs, FIES artifacts are written to FIES-outputs below this folder.

Value

The session, invisibly.

read_mics_tabulation	<i>Read one sheet from a MICS Plus tabulation plan</i>
----------------------	--

Description

Read one sheet from a MICS Plus tabulation plan

Usage

```
read_mics_tabulation(session, path_tab_excel, sheet)
```

Arguments

session	A session created by <code>mics_session()</code> .
path_tab_excel	Path to the Excel tabulation plan.
sheet	Sheet name or index.

Value

The parsed tabulation-plan context.

read_previous_mics_table

Read a previously generated worksheet for comparison

Description

Read a previously generated worksheet for comparison

Usage

```
read_previous_mics_table(path, sheet, skip = 3)
```

Arguments

path	Excel workbook path.
sheet	Sheet name or index.
skip	Number of header rows to skip.

Value

A cleaned table.

run_app

Run the micsPlusTableR Shiny application

Description

Starts the Shiny application bundled with micsPlusTableR. The Download Files tab saves survey plans and preparation files to the current project by default; its destination can be changed in the tab.

Usage

```
run_app(
  host = "127.0.0.1",
  port = getOption("shiny.port"),
  launch.browser = interactive(),
  output_dir = NULL,
  ...
)
```

Arguments

host	Host interface passed to shiny::runApp().
port	Port passed to shiny::runApp(); NULL selects a free port.
launch.browser	Whether to open the application in a browser.
output_dir	Default directory for generated Excel workbooks. When NULL, the app creates micsPlusTableR-output in the project root detected by here::here().
...	Additional arguments passed to shiny::runApp().

Value

Invisibly returns the value from `shiny::runApp()`.

survey_choices	<i>Maintain survey and wave choices</i>
----------------	---

Description

The plain UTF-8 CSV file ‘inst/extdata/survey_choices.csv’ in the package source controls the application’s survey and wave selectors. Edit this source file directly before committing to GitHub or building the package.

Columns

Each row represents one available country, period, and wave combination. All four columns are read as character strings:

country Full country name, for example Jamaica.

country_code Exactly three uppercase letters, for example JAM.

period Survey period as text, for example 2023–24.

wave Wave label, for example Wave 1.

Keep the column names. The file must have at least one row, with no empty fields or duplicate country/period/wave combinations. All waves of the same country and period must use the same country code. Leading and trailing whitespace is removed. Row order determines selector order and defaults.

Editing and installation

Use a text editor or a spreadsheet editor that preserves the period and wave columns as text. Save as UTF-8 CSV. From the package root, run `source("run_dev.R")` to load and launch the development source. Stop and restart the app after saving changes.

The CSV requires no conversion to an R data object.

R installs ‘inst/extdata’ files into the package’s ‘extdata’ directory. For installed packages, reinstall the updated source and restart the app. Direct edits to an installed copy are replaced by re-installation or upgrades. The selected code, full period, and wave also contribute to output filenames and long-format metadata.

See Also

[run_app, micsPlusTableR-package](#)

Examples

```
path <- system.file("extdata", "survey_choices.csv", package = "micsPlusTableR")
choices <- read.csv(path, colClasses = "character", fileEncoding = "UTF-8-BOM")
choices
```

tabulate_mics_table	<i>Tabulate a MICS Plus table outside the application</i>
---------------------	---

Description

Tabulate a MICS Plus table outside the application

Usage

```
tabulate_mics_table(session, path_tab_excel = NULL, sheet = NULL)
```

Arguments

session	A prepared tabulation session.
path_tab_excel	Optional tabulation-plan path. Supply with sheet to read a new table before tabulating.
sheet	Optional sheet name or index.

Value

A long-format data frame containing one row per calculated cell.

validate_sdmx_compatible	<i>Validate SDMX-compatible MICS long-format data</i>
--------------------------	---

Description

Checks the complete 35-variable contract and the unique observation key: SURVEY_ID, TABLE_ID, row_order, column_order, and TIME_PERIOD.

Usage

```
validate_sdmx_compatible(data)
```

Arguments

data	A data frame produced by as_sdmx_compatible() .
------	---

Value

data, invisibly.

write_mics_footnotes	<i>Add standard footnotes to a tabulation workbook</i>
----------------------	--

Description

Add standard footnotes to a tabulation workbook

Usage

```
write_mics_footnotes(session, destination, sheet = NULL, table = NULL)
```

Arguments

session	A tabulation session.
destination	Existing destination workbook.
sheet	Sheet name or index; defaults to the current sheet.
table	Cell results; defaults to the latest result.

Value

The destination path, invisibly.

write_mics_table	<i>Write cell results to a tabulation workbook</i>
------------------	--

Description

Write cell results to a tabulation workbook

Usage

```
write_mics_table(
  session,
  destination,
  sheet = NULL,
  table = NULL,
  formatted = FALSE,
  drop_n_unw = FALSE
)
```

Arguments

session	A tabulation session with a loaded plan.
destination	Existing destination workbook.
sheet	Sheet name or index; defaults to the current sheet.
table	Cell results; defaults to the latest result.
formatted	Whether to write formatted display values.
drop_n_unw	Whether to omit unweighted-count columns when formatting.

Value

The destination path, invisibly.

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